

Detailed information on the XAI and UQ models used.

Dataset

This study uses two medical datasets – Lung and Colorectal cancer datasets. Both were derived from the Singapore Chinese Health Study (SCHS) [1] which studies the effects of diet, environment and genetics on the development of cancer and other chronic disease in the ethnic Chinese population in Singapore.

The Lung cancer dataset consists of 23622 instances. Each instance describes the condition of a patient using 5 categorical and 16 continuous features and a class label describing if the patient has Lung cancer. The Colorectal cancer dataset consists of 22901 instances. Each instance describes the condition of a patient using 5 categorical and 12 continuous features and a class label describing if the patient has Colorectal cancer.

Both datasets were pre-processed with the following steps: Patients with missing values were dropped, continuous features were min-max normalised while categorical features were one-hot encoded. We randomly divided the dataset into training (80%), validation (10%) and test sets (10%) for model training, tuning and evaluation respectively. As both datasets are class-imbalanced, we oversampled the cancer instances in the training set with SMOTE [2].

We selected four instances from the test sets of each dataset as examples for the survey. These four instances represent all combinations of the class labels (Normal and Cancer) and correct or incorrect predictions.

Table 2. Description of Lung and Colorectal cancer datasets and model parameters for each model.

Data Subset	Cancer Proportion (%)	Size		Feature Selection		Hyperparameters	
		Before Dropping	After Dropping	Features	Selected Features	Trees	Max Depth
Colorectal	2.58	22901	20689	17	7	50	5
Lung	1.90	23622	17464	21	15	50	10

A.1.1 AI Model

We trained a Random Forest [3] model on each dataset. Random Forest is an ensemble method which aggregates the predictions of multiple decision trees improving its accuracy and robustness to overfitting, which has made it a widely studied model for medical diagnosis [4, 5]. Each decision tree predicts the class of a new instance by sorting the instance into a leaf node based on its features. The majority class in the leaf node determines the class prediction for that instance. To make the final prediction, the random forest model aggregates predictions from each decision tree through majority voting.

To improve prediction performance, we employed feature selection. We estimated the mutual information [6] between each feature and the class label. A high mutual information indicates that the feature is important for prediction. We incrementally added features to the feature set in descending order of importance and evaluated models trained on the feature set on the validation set using AUC (Receiver Operating Characteristic Area Under the Curve). We selected the feature set with the highest AUC. To determine the hyperparameters for the random forest model (Number of trees and max depth), we used grid search $n_trees\{50, 100, 200, 300, 400, 500, 600, 700\}$ and $max_depth\{5, 10, 20, 30, 40\}$ and selected the hyperparameters with the highest AUC on the validation set. Table 2 indicates the number of selected features and hyperparameters for each dataset. Table 3 displays the performance of each model on each data subset.

Table 3. Model prediction performance.

Data Subset	Lung Cancer		Colorectal Cancer	
	AUC	Accuracy	AUC	Accuracy
Train	0.968	0.829	0.999	0.862
Valid	0.816	0.818	0.623	0.833
Test	0.734	0.802	0.632	0.831

A.1.2 XAI Method

We utilised SHAP [7] (SHapley Additive exPlanations), a widely used XAI technique in medical diagnosis [8], to derive feature attribution scores from each model for each instance in the survey. Each instance has as many SHAP feature attribution scores as the number of features in the dataset. The sign of the score indicates how the feature affects prediction – negative values indicate that the feature negatively contributes to the prediction and vice versa. The magnitude of the score indicates how strongly the feature attributes prediction – high feature attribution values indicate that the feature is important for prediction. To improve comprehensibility, we have displayed the feature attribution score on a bar chart in the survey and indicated risk factors in red and protective factors in green together with a legend.

A.1.3 UQ Method

Besides providing respondents with the overall accuracy of each model (Indirect Uncertainty), we also provide uncertainty score and bounds (Direct Uncertainty) [9] derived from the Entropy [10, 11] of prediction probabilities. Entropy is the most accessible uncertainty estimate for classification problems. It is computed as the Shannon’s entropy (Equation 1) of ML prediction probabilities $p(x)$.

$$h = -\sum_i p_i(x) \log_2 p_i(x), i \in \{0, 1\} \quad (1)$$

To improve the interpretability of our uncertainty score, we further processed entropy to yield an uncertainty percentage. To derive this value, we define the incorrect prediction probability e as one minus the probability assigned to the predicted class, $p_y(x)$, where y denotes the predicted label ($y = 1$, for cancer; $y = 0$, for normal):

$$e = 1 - p_y(x).$$

Next, we fit a multivariate Gaussian distribution on the entropy h and incorrect prediction probabilities e of the validation set:

$$\begin{bmatrix} e \\ h \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_e \\ \mu_h \end{bmatrix}, \begin{bmatrix} \Sigma_{ee} & \Sigma_{eh} \\ \Sigma_{he} & \Sigma_{hh} \end{bmatrix} \right).$$

With this multivariate Gaussian distribution, we can compute the conditional mean and variance of incorrect prediction probability e given the entropy of this new instance, $\mu(e|h)$ and $\sigma^2(e|h)$.

$$\mu_{e|h} = \mu_e + \Sigma_{eh} \Sigma_{hh}^{-1} (h - \mu_h), \quad \sigma_{e|h}^2 = \Sigma_{ee} - \Sigma_{eh} \Sigma_{hh}^{-1} \Sigma_{he}.$$

Since the incorrect prediction probability e is bounded between 0 and 1, we further truncate the conditional Gaussian distribution such that $0 \leq e \leq 1$, yielding a modified conditional mean and variance. Specifically, the truncated conditional distribution is given by:

$$e | h, \quad 0 \leq e \leq 1 \sim \mathcal{N}_{[0,1]}(\mu_{e|h}, \sigma_{e|h}^2).$$

Let $\alpha = \frac{0 - \mu_{e|h}}{\sigma_{e|h}}$ and $\beta = \frac{1 - \mu_{e|h}}{\sigma_{e|h}}$, the modified conditional mean is then:

$$\tilde{\mu}_{e|h} = \mu_{e|h} - \sigma_{e|h} \left(\frac{\phi(\beta) - \phi(\alpha)}{\Phi(\beta) - \Phi(\alpha)} \right),$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ denote the probability density function (PDF) and cumulative distribution function (CDF) of the standard normal distribution, respectively. The modified conditional variance is:

$$\tilde{\sigma}_{e|h}^2 = \sigma_{e|h}^2 \left[1 - \left(\frac{\beta\phi(\beta) - \alpha\phi(\alpha)}{\Phi(\beta) - \Phi(\alpha)} \right) - \left(\frac{\phi(\beta) - \phi(\alpha)}{\Phi(\beta) - \Phi(\alpha)} \right)^2 \right].$$

The modified conditional mean $\mu(e|h)$ serves as the uncertainty probability of the instance. We also compute the lower and upper bound of uncertainty by adding and subtracting 1.96 standard deviations from the modified conditional mean:

$$[\tilde{\mu}_{e|h} - 1.96 \tilde{\sigma}_{e|h}^2, \tilde{\mu}_{e|h} + 1.96 \tilde{\sigma}_{e|h}^2].$$

These bounds provide an approximately 95% interval around the estimated uncertainty probability.

Table 8 evaluates the performance of our uncertainty score on each data subset. We observe high correlations between incorrect prediction probabilities and our uncertainty scores across both datasets; alongside empirical coverage values close to the expected coverage of 0.95.

Table 8. Uncertainty quantification performance.

Metric		Correlation		Coverage
Data Subset	Lung	Colorectal	Lung	Colorectal
Train	0.755	0.845	0.945	0.968
Valid	0.745	0.780	0.942	0.956
Test	0.713	0.775	0.929	0.948

A.1.3 References

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